

SECTION 03 30 00

CAST-IN-PLACE CONCRETE

Riverside Mixed-Use Development
Project No. RMD-2025-001

PART 1 - GENERAL

1.01 REFERENCES

- ACI 301 - Specifications for Structural Concrete
- ACI 318 - Building Code Requirements for Structural Concrete
- ASTM C31 - Making and Curing Concrete Test Specimens in the Field
- ASTM C39 - Compressive Strength of Cylindrical Concrete Specimens
- ASTM C94 - Specification for Ready-Mixed Concrete
- ASTM C150 - Specification for Portland Cement
- ASTM C260 - Air-Entraining Admixtures for Concrete

1.02 SUBMITTALS

- A. Mix design submittals for each concrete class, including 28-day compressive strength test results from an independent testing laboratory.
- B. Material certificates for cement, aggregates, admixtures, and reinforcing steel.
- C. Concrete placement plan for each major pour, including sequence, joint locations, and curing method.

1.03 QUALITY ASSURANCE

- A. Concrete supplier shall be certified by the National Ready Mixed Concrete Association (NRMCA).
- B. Testing laboratory shall be accredited per ASTM C1077.
- C. Finisher qualification: minimum 5 years experience on similar commercial projects.

1.04 DELIVERY, STORAGE, AND HANDLING

- A. Deliver concrete in truck mixers complying with ASTM C94.
- B. Maintain delivery tickets for each load showing batch time, water added, admixtures, and truck number.
- C. Reject any concrete that has exceeded 90-minute haul time or 300 drum revolutions.

PART 2 - PRODUCTS

2.01 CONCRETE MIX DESIGN

Class	f'c (PSI)	Slump (in)	Air	Max W/C	Application
A	5,000	4 +/- 1	5-7%	0.40	Foundations, SOG
B	4,000	4 +/- 1	5-7%	0.45	Elevated slabs, beams
C	6,000	4 +/- 1	5-7%	0.35	Columns, shear walls
D	3,000	6 +/- 1	4-6%	0.50	Fill, non-structural

2.02 MATERIALS

A. Portland Cement: ASTM C150, Type I/II. Manufacturer: Lehigh Hanson or approved equal.

B. Aggregates: ASTM C33, Size No. 57 coarse aggregate. Fine aggregate: natural sand, FM 2.5-3.0.

C. Water: Potable, free from oils, acids, and organic matter.

D. Admixtures: Air-entraining ASTM C260 MasterAir AE 200; Water-reducing ASTM C494 Type A MasterGlenium 7920; Accelerating (cold weather) ASTM C494 Type C.

2.03 REINFORCING STEEL

A. Deformed bars: ASTM A615, Grade 60.

B. Welded wire reinforcement: ASTM A185.

C. Bar supports: CRSI Class 1 interior, Class 2 exterior.

D. Lap splices per ACI 318 Chapter 25.

PART 3 - EXECUTION

3.01 PREPARATION

- A. Verify formwork dimensions, alignment, and adequacy of shoring before placing concrete.
- B. Moisten subgrade and forms before placement. Remove standing water.

3.02 PLACEMENT

- A. Place concrete within 90 minutes of batching, or within 300 drum revolutions.
- B. Maximum free-fall height: 5 feet. Use tremie or pump for greater heights.
- C. Consolidate with internal vibrators at 18-inch spacing.
- D. Cold weather: maintain concrete temperature above 50 deg F for minimum 7 days.

3.03 CURING

- A. Begin curing immediately after finishing operations.
- B. Wet cure for minimum 7 days using water-saturated burlap and polyethylene sheeting.
- C. Curing compound alternative: ASTM C309, Type 1, Class B.

3.04 FIELD QUALITY CONTROL

- A. Sample and test per ASTM C31: minimum one set of 4 cylinders per 50 CY or per day of placement.
- B. Slump test per ASTM C143 for each truckload.
- C. Air content test per ASTM C231 for each truckload.
- D. Compressive strength tests per ASTM C39 at 7 and 28 days.

END OF SECTION 03 30 00